Logarithmic Functions & Properties

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Converting between exponential and logarithmic form

Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.

$2^5 = 32 \Leftrightarrow \log_2 32 = 5$. $\log_3 81 = 4 \Leftrightarrow 3^4 = 81$.

2 [Foundational] • Target: Converting between exponential and logarithmic form

Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.

$\log_5 125 = 3$ since $5^3 = 125$. $\log_2 \frac{1}{8} = -3$ since $2^{-3} = \frac{1}{8}$.

3 [Foundational] • Target: Properties of logarithms

Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.

$\log\left(\frac{x^3\sqrt{y}}{z^2}\right) = \log(x^3) + \log(y^{1/2}) - \log(z^2) = 3\log x + \frac{1}{2}\log y - 2\log z$.

4 [Foundational] • Target: Properties of logarithms

Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.

$2\log x - \frac{1}{3}\log y + \log z = \log(x^2) - \log(y^{1/3}) + \log z = \log\left(\frac{x^2 z}{y^{1/3}}\right)$.

5 [Foundational] • Target: Properties of logarithms

Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

$\log_7 50 = \frac{\ln 50}{\ln 7} = \frac{3.912}{1.946} \approx 2.010$.

6 [Foundational] • Target: Visualizing logarithmic functions

The graph shows $f(x) = \ln(x)$, the natural logarithm.

a State the domain and the vertical asymptote shown in the graph. Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \to 0^+$.

b State the x-intercept shown. x-intercept at $(1, 0)$, since $\ln(1) = 0$.

(a) Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \to 0^+$.; (b) x-intercept at $(1, 0)$, since $\ln(1) = 0$.

7 [Foundational] • Target: Solving exponential equations using logarithms

Solve $5^x = 40$ for x, to three decimal places.

Taking $\log$ of both sides: $x\log 5 = \log 40$, so $x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving exponential equations using logarithms

Solve $3(2)^{x-1} = 45$ for x, to three decimal places.

$2^{x-1} = 15$. Taking $\log$: $(x-1)\log 2 = \log 15$, so $x - 1 = \frac{\log 15}{\log 2} \approx 3.907$, giving $x \approx 4.907$.

9 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_2(x+3) = 4$ for x, and check that the solution keeps the argument positive.

$x + 3 = 2^4 = 16$, so $x = 13$. Check: $x + 3 = 16 > 0$ ✓. Solution: $x = 13$.

10 [Intermediate] • Target: Solving logarithmic equations

Solve $\log(x) + \log(x-3) = 1$ for x, checking both solutions against the domain.

Combine: $\log[x(x-3)] = 1$, so $x(x-3) = 10$: $x^2 - 3x - 10 = 0$, $(x-5)(x+2) = 0$: $x = 5$ or $x = -2$. Since $x > 0$ and $x - 3 > 0$ are required (domain of the original logs), $x = -2$ is rejected. Solution: $x = 5$.

11 [Intermediate] • Target: Applications: the pH scale and decibels

The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.

$\text{pH} = -\log(3.2 \times 10^{-5}) \approx -(-4.495) = 4.49$.

12 [Intermediate] • Target: Applications: the pH scale and decibels

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

Difference $= 10\log(1000) - 10\log(1) = 10\log(1000) = 10(3) = 30$ decibels.

13 [Intermediate] • Target: Transformations of logarithmic functions

Describe how the graph of $g(x) = \ln(x+2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.

$g(x) = \ln(x+2) - 3$ shifts $f(x) = \ln(x)$ left 2 units and down 3 units. The vertical asymptote shifts from $x = 0$ to $x = -2$, and the domain becomes $x > -2$.

14 [Intermediate] • Target: Transformations of logarithmic functions

Find the domain of $h(x) = \log(4 - x^2)$.

Require $4 - x^2 > 0$: $x^2 < 4$, so $-2 < x < 2$. Domain: $(-2, 2)$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Solving equations that combine exponentials and logarithms** • **Cross-Lesson: 1.01 Polynomial End Behavior & Zeros**

Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.

Let $u = e^x$: $u^2 - 5u + 6 = 0$, so $(u - 2)(u - 3) = 0$, giving $u = 2$ or $u = 3$. Since $u = e^x > 0$ for all real x, both are valid: $e^x = 2 \Rightarrow x = \ln 2$, or $e^x = 3 \Rightarrow x = \ln 3$.

16 **[Challenging]** • **Target: Solving equations that combine exponentials and logarithms** • **Cross-Lesson: 1.02 Rational Functions & Asymptotes**

Solve $\log_2(x) + \log_2(x - 2) = 3$ for x, checking the domain.

Combine: $\log_2[x(x - 2)] = 3$, so $x(x - 2) = 2^3 = 8$: $x^2 - 2x - 8 = 0$, $(x - 4)(x + 2) = 0$: $x = 4$ or $x = -2$. Domain requires $x > 0$ and $x - 2 > 0$ (i.e. $x > 2$), so $x = -2$ is rejected. Solution: $x = 4$.

17 **[Challenging]** • **Target: Comparing logarithmic growth to other function families** • **Cross-Lesson: 1.03 Rates of Change & Concavity**

Explain why logarithmic functions grow much more slowly than linear functions for large x, and give a numerical comparison of $\log(x)$ and x at $x = 1000$.

Logarithms compress large inputs: each tenfold increase in x adds only 1 to $\log(x)$, while a linear function adds proportionally to the increase in x itself. At $x = 1000$: $\log(1000) = 3$, while the linear function $y = x$ gives 1000 — the logarithm is vastly smaller, illustrating its extremely slow long-run growth.

18 **[Challenging]** • **Target: Modeling with logarithms: the Richter scale** • **Cross-Lesson: 2.01 Exponential Functions & Growth Models**

Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

$7 - 5 = \log\left(\frac{I_7}{I_0}\right) - \log\left(\frac{I_5}{I_0}\right) = \log\left(\frac{I_7}{I_5}\right) = 2$, so $\frac{I_7}{I_5} = 10^2 = 100$. The magnitude-7 earthquake is 100 times more intense.

19 **[Challenging]** • **Target: Modeling with logarithms: the Richter scale** • **Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

If an earthquake's intensity is $10{,}000$ times a reference intensity I_0, find its Richter magnitude.

$M = \log(10000) = \log(10^4) = 4$.

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20 [Challenging] • Target: Applications: the pH scale and decibels • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Solve the logarithmic equation $\log_4(x+6)+\log_4(x-6)=3$, checking for extraneous roots.

Step 1: Domain restriction: We require $x+6>0$ and $x-6>0$, so $x>6$. Step 2: Condense logarithms using the product rule: $\log_4((x+6)(x-6))=3 \Rightarrow x^2-36=4^3=64$. Step 3: Solve for x and filter by domain: $x^2=100 \Rightarrow x=\pm 10$. Reject $x=-10$ since $-10 ot > 6$. The unique valid solution is **$x=10$**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between exponential and logarithmic form

Evaluate $\log_2(2)$ without a calculator.

a. 1 ✓ **b.** -1

c. 0 **d.** 2

$\log_2(2)=1$ since $2^1=2$.

22 [Foundational] • Target: Properties of logarithms

Evaluate $\log_6(36)$ without a calculator.

a. 2 ✓ **b.** 1

c. -2 **d.** 3

$\log_6(36)=2$ since $6^2=36$.

23 [Foundational] • Target: Visualizing logarithmic functions

Evaluate $\log_4(16)$ without a calculator.

a. 3 **b.** 2 ✓

c. -2 **d.** 1

$\log_4(16)=2$ since $4^2=16$.

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24 [Foundational] • Target: Solving exponential equations using logarithms

Evaluate $\log_4(64)$ without a calculator.

a. -3

b. 3 ✓

c. 4

d. 2

$\log_4(64) = 3$ since $4^3 = 64$.

25 [Foundational] • Target: Solving logarithmic equations

Evaluate $\log_4(4)$ without a calculator.

a. 2

b. 0

c. -1

d. 1 ✓

$\log_4(4) = 1$ since $4^1 = 4$.

26 [Foundational] • Target: Applications: the pH scale and decibels

Evaluate $\log_2\left(\frac{1}{2}\right)$ without a calculator.

a. 0

b. 1

c. -2

d. -1 ✓

$\log_2\left(\frac{1}{2}\right) = -1$ since $2^{-1} = \frac{1}{2}$.

27 [Foundational] • Target: Transformations of logarithmic functions

Evaluate $\log_4\left(\frac{1}{256}\right)$ without a calculator.

a. -3

b. -5

c. -4 ✓

d. 4

$\log_4\left(\frac{1}{256}\right) = -4$ since $4^{-4} = \frac{1}{256}$.

28 [Foundational] • Target: Solving equations that combine exponentials and logarithms

Solve $\log_6(x) = 3$ for x.

a. 215

b. 217

c. 216 ✓

d. 218

$x = 6^3 = 216$, so $x = 216$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Comparing logarithmic growth to other function families**

Solve $\log_5(x-5)=3$ for x.

a. 129

b. 130 ✓

c. 120

d. 131

$x-5=5^3=125$, so $x=125+5=130$.

30 **[Intermediate]** • **Target: Modeling with logarithms: the Richter scale**

Solve $\log_4(x+9)=1$ for x.

a. 13

b. -4

c. -5 ✓

d. -6

$x+9=4^1=4$, so $x=4-9=-5$.

31 **[Intermediate]** • **Target: Definition of Logarithms and Inverses**

Solve $\log_3(x+4)=2$ for x.

a. 13

b. 4

c. 5 ✓

d. 6

$x+4=3^2=9$, so $x=9-4=5$.

32 **[Intermediate]** • **Target: Logarithmic Properties & Condensing Expressions**

Solve $\log_6(x-3)=1$ for x.

a. 8

b. 9 ✓

c. 10

d. 3

$x-3=6^1=6$, so $x=6+3=9$.

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33 [Intermediate] • Target: Solving Exponential and Logarithmic Equations

Solve $\log_2(x+7)=1$ for x.

a. -5 ✓

b. 9

c. -4

d. -6

$x+7=2^1=2$, so $x=2-7=-5$.

34 [Intermediate] • Target: Logarithmic Scales and Applications

Solve $\log_6(x-6)=2$ for x.

a. 42 ✓

b. 43

c. 30

d. 41

$x-6=6^2=36$, so $x=36+6=42$.

35 [Intermediate] • Target: Converting between exponential and logarithmic form

Solve $9^x=165$ for x, to three decimal places.

a. 2.824

b. 2.424

c. 2.224

d. 2.324 ✓

Taking $\log$ of both sides: $x\log 9=\log 165$, so $x=\frac{\log 165}{\log 9}\approx 2.324$.

36 [Intermediate] • Target: Properties of logarithms

Solve $6^x=90$ for x, to three decimal places.

a. 2.411

b. 2.511 ✓

c. 2.611

d. 3.011

Taking $\log$ of both sides: $x\log 6=\log 90$, so $x=\frac{\log 90}{\log 6}\approx 2.511$.

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37 [Intermediate] • Target: Visualizing logarithmic functions

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^6 times more intense than another, find the difference in their decibel levels.

a. 50

b. 70

c. 60 ✓

d. 6

Difference $= 10\log(10^6) - 10\log(1) = 10(6) = 60$ decibels.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^2 times more intense than another, find the difference in their decibel levels.

a. 20 ✓

b. 10

c. 2

d. 30

Difference $= 10\log(10^2) - 10\log(1) = 10(2) = 20$ decibels.

39 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Solve $\log_{21}(x) + \log_{21}(x-4) = 1$ for x, checking both solutions against the domain.

a. 8

b. -3

c. 7 ✓

d. 4

Combine: $\log_{21}[x(x-4)] = 1$, so $x(x-4) = 21$: $x^2 - 4x - 21 = 0$, $(x-(7))(x-(-3)) = 0$: $x = 7$ or $x = -3$. Since $x > 0$ and $x - 4 > 0$ are required, $x = -3$ is rejected. Solution: $x = 7$.

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40 [Challenging] • Target: Applications: the pH scale and decibels • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Solve $\log_6(x) + \log_6(x-5) = 1$ for x, checking both solutions against the domain.

a. 6 ✓

b. 7

c. -1

d. 5

Combine: $\log_6[x(x-5)] = 1$, so $x(x-5) = 6$: $x^2 - 5x - 6 = 0$, $(x-(6))(x-(-1)) = 0$: $x = 6$ or $x = -1$. Since $x > 0$ and $x - 5 > 0$ are required, $x = -1$ is rejected. Solution: $x = 6$.

41 [Challenging] • Target: Definition of Logarithms and Inverses • Cross-Lesson: 4.01 Inverse & Composite Functions

Evaluate $\log_4\left(\frac{1}{64}\right)$.

a. -3 ✓

b. 3

c. $-\frac{1}{3}$

d. 16

Let $y = \log_4(1/64) \Leftrightarrow 4^y = \frac{1}{64} = 4^{-3} \Rightarrow y = -3$.

42 [Challenging] • Target: Definition of Logarithms and Inverses • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Find the inverse function of $g(x) = 3 + 2\ln(x-1)$.

a. $g^{-1}(x) = e^{(x-3)/2} + 1$ ✓

b. $g^{-1}(x) = e^{(x-3)/2} - 1$

c. $g^{-1}(x) = 2e^{x-3} + 1$

d. $g^{-1}(x) = e^{2(x-3)} + 1$

Set $y = 3 + 2\ln(x-1) \Rightarrow y - 3 = 2\ln(x-1) \Rightarrow \ln(x-1) = \frac{y-3}{2} \Rightarrow x - 1 = e^{(y-3)/2} \Rightarrow x = e^{(y-3)/2} + 1$.

43 [Challenging] • Target: Logarithmic Properties & Condensing Expressions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Using the change of base formula, express $\log_5 12$ in terms of natural logarithms.

a. $\frac{\ln 12}{\ln 5}$ ✓

b. $\frac{\ln 5}{\ln 12}$

c. $\ln 12 - \ln 5$

d. $\ln\left(\frac{12}{5}\right)$

The change of base formula states $\log_b a = \frac{\log_c a}{\log_c b}$. For base e, $\log_5 12 = \frac{\ln 12}{\ln 5}$.

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44 [Challenging] • Target: Solving Exponential and Logarithmic Equations • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve for x: $\log_2(x) + \log_2(x-2) = 3$.

a. 4 ✓

b. 4 and -2

c. 2

d. 8

Combine: $\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since the domain requires $x > 2$, $x = -2$ is extraneous. The only valid solution is $x = 4$.

45 [Challenging] • Target: Logarithmic Scales and Applications • Cross-Lesson: 1.02 Rational Functions & Asymptotes

The pH of a chemical solution is given by $\text{pH} = -\log[\text{H}^+]$. If the hydrogen ion concentration is 1.0×10^{-5} M, what is the pH?

a. 5.0 ✓

b. -5.0

c. 1.0

d. 9.0

$\text{pH} = -\log(10^{-5}) = -(-5) = 5.0$.

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